Running Local Large Language Models on Personal Hardware

Individuals and organizations today rely heavily on artificial intelligence tools like ChatGPT and Claude. When a user submits a prompt, external data centers compute the response. This cloud setup is powerful but presents two major risks. Sending private files or proprietary data to an external provider creates security vulnerabilities, and frequent use leads to expensive usage costs.

A local Large Language Model (LLM) solves both problems by running the artificial intelligence directly on private hardware. Tech companies are now releasing open source, or open weight (where the model's trained parameters are publicly accessible for anyone to download and use), models directly to the public, like the Llama models from Meta and the Gemma series from Google. Because anyone can download and run these open weight models on their own machines, they are also commonly called local LLMs. By hosting them locally, a workstation performs every calculation offline. This keeps data entirely private and eliminates cloud usage fees completely. This document gives a breakdown of how local and cloud processing compare, the hardware required, and how unified memory makes enterprise-tier AI possible on a single machine.

Cloud versus Local Prompt Processing

Cloud Processing: Platforms like ChatGPT host their models across massive server clusters. Users are renting computational power from an external provider. Operations require an active internet connection, and costs scale exponentially as usage increases.

Local Processing: Running an AI locally moves the model directly onto internal storage. The computer loads the model into its own memory. When given a prompt, the local hardware executes the generation cycle entirely offline and displays the text directly on the screen.

Hardware Requirements and Model Tiers

Running a language model locally relies entirely on the physical components of the computer. The capability and intelligence of a model depend on its size, which is measured in parameters. More parameters result in better reasoning, but running larger models requires much more hardware power.

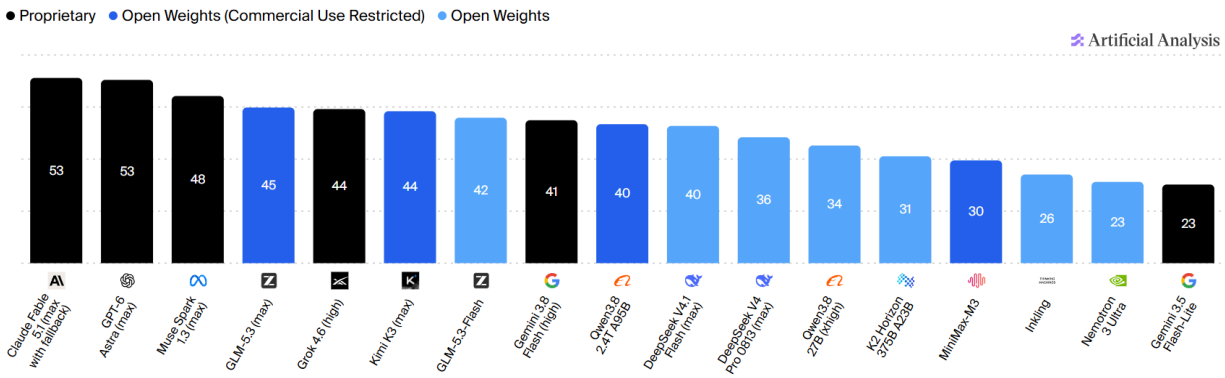


Figure 1: Artificial Analysis Intelligence Index, artificialanalysis.ai. This index aggregates multiple benchmarks to score intelligence. Black bars represent cloud models, and blue bars indicate open weight local models.

Memory (VRAM) Limits: The entire file containing the learned parameters must fit into active memory. Language models operate best inside Video RAM located directly on a graphics card. If a downloaded model file requires eight gigabytes of space, the hardware must have at least eight gigabytes of Video RAM to open it.

Consumer Hardware: Because model intelligence is tied to memory capacity, different hardware setups allow you to run different tiers of artificial intelligence. Standard consumer laptops or desktop computers with eight to sixteen gigabytes of Video RAM easily run smaller models. These models are highly capable for everyday coding or writing tasks for individual users.

Workstation Class AI: Running massive advanced models requires specialized hardware. Devices like the Apple Mac Studio or the NVIDIA DGX Spark are built specifically for this purpose. Instead of a standard graphics card, these compact workstations use massive pools of unified memory that often exceed one hundred gigabytes. Unified memory acts as a massive, shared pool of high-speed memory that both the CPU and GPU can access simultaneously, eliminating the need to copy data back and forth.These workstation devices allow solo developers and engineering teams to achieve cloud level intelligence and speed entirely locally. Users can run massive models natively right on their desks without needing a full data center.

Running a local Large Language Model gives individuals and organizations the full utility of modern AI while preserving total data ownership and eliminating cloud fees. By understanding how model size connects to dedicated memory and workstation hardware, professionals can better evaluate the equipment needed to bring artificial intelligence securely in house.